

Climate Change

Aleph Jimenez

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BACKGROUND

The planet's surface temperature is increasing as greenhouse gas emissions increase, and this global warming and carbon cycle disruption is wreaking havoc on natural systems. Living systems that depend on current temperature, weather, currents and carbon balance are jeopardized, and human society will be forced to contend with widespread economic, social, political and environmental damage as the temperature continues to rise. We examine the relationship between global temperature changes, greenhouse gases and human carbon emissions using time series of actual atmospheric and ice core measurements from the National Oceanic and Atmospheric Administration (NOAA) and Carbon Dioxide Information Analysis Center (CDIAC).

A major determinant of Earth's temperature is the greenhouse effect. Many gases (e.g. CO₂, CH₄, N₂O) trap heat and reflect it towards the surface, preventing heat from escaping the atmosphere. The greenhouse effect is vital in keeping Earth at a warm enough temperature to sustain liquid water and life; however, changes in greenhouse gas levels can alter the temperature balance of the planet.

Greenhouse gases have changed over the course of human history, but how has CO₂ varied over a longer time scale?

One way to differentiate natural CO₂ oscillations from today's manmade CO₂ spike is by examining the rate of change of CO₂. The planet is affected not only by the absolute concentration of CO₂ but also by its rate of change. When the rate of change is slow, living and nonliving systems have time to adapt to new temperature and gas levels, but when the rate of change is fast, abrupt differences can overwhelm natural systems. How does the pace of natural CO₂ change differ from the current rate of change?

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.2      v readr      2.1.4
## v forcats    1.0.0      v stringr   1.5.0
## v ggplot2    3.4.2      v tibble    3.2.1
## v lubridate  1.9.2      v tidyr     1.3.0
## v purrr      1.0.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
##
## Attaching package: 'gridExtra'
##
##
## The following object is masked from 'package:dplyr':
##
##   combine
##
```

```
##
## ##
## ## Amelia II: Multiple Imputation
## ## (Version 1.8.1, built: 2022-11-18)
## ## Copyright (C) 2005-2024 James Honaker, Gary King and Matthew Blackwell
## ## Refer to http://gking.harvard.edu/amelia/ for more information
## ##
##
## Registered S3 method overwritten by 'GGally':
##   method from
##   +.gg      ggplot2

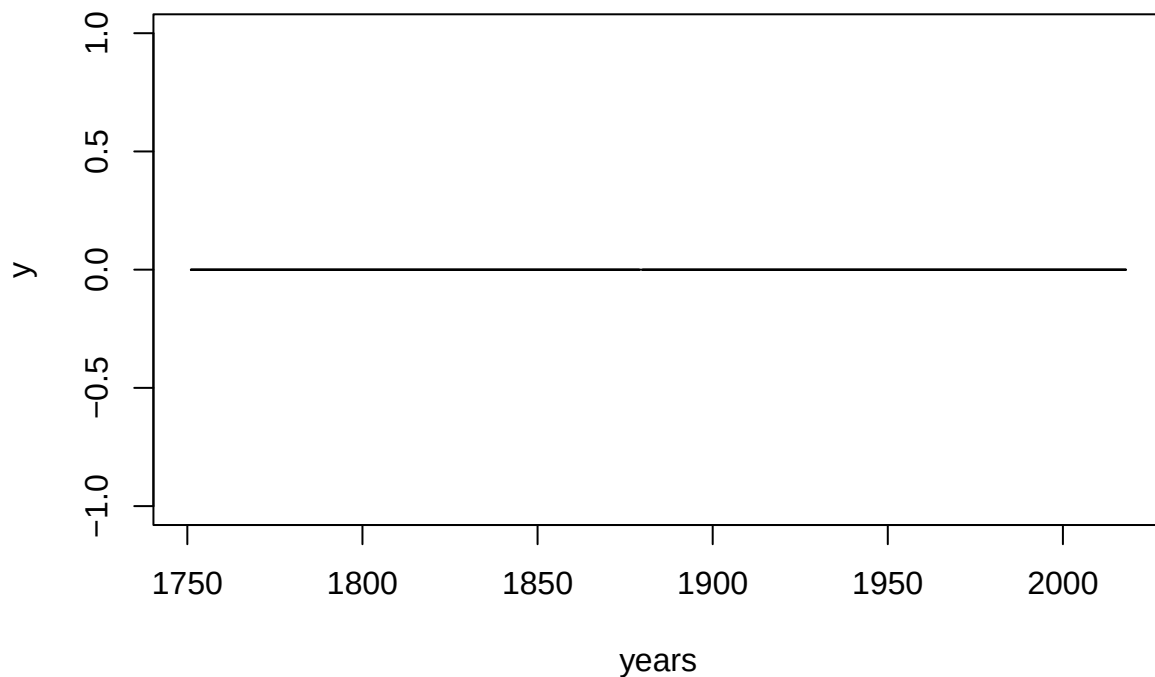
## Warning: package 'dslabs' was built under R version 4.3.3
```

EXPLORATORY

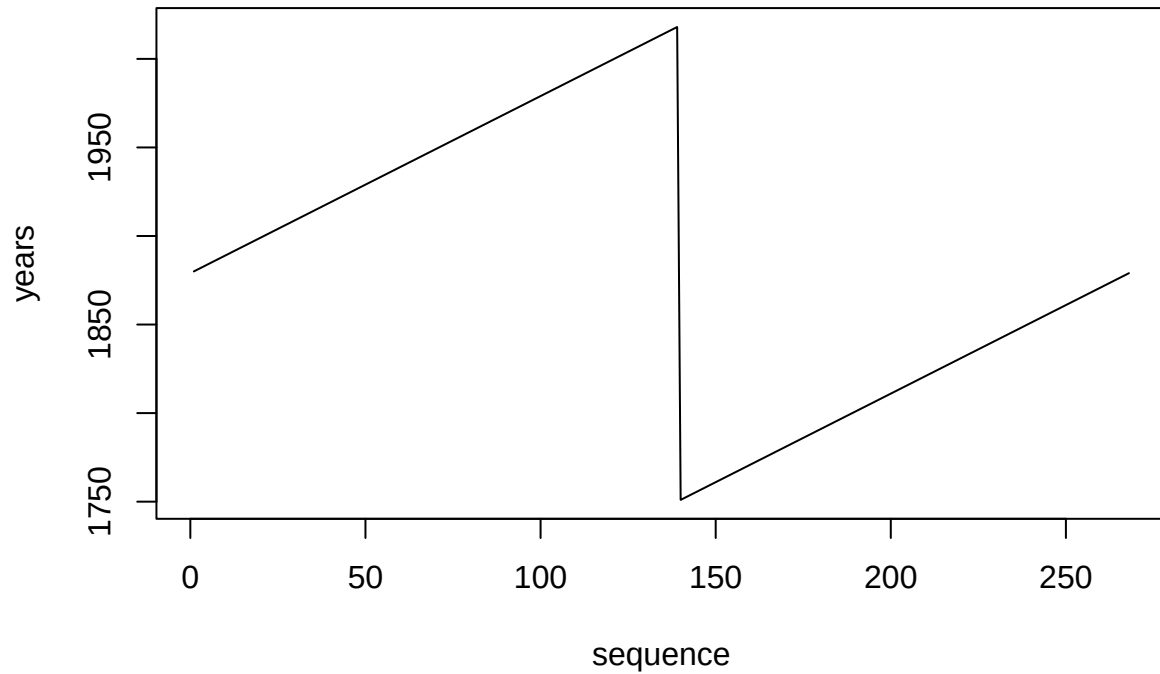
1. TEMP_CARBON

Let's check that time vector doesn't have gaps.

```
years <- temp_carbon$year
y <- integer(length(temp_carbon$year))
plot(years, y, "l")
```



```
sequence <- seq(1,268)
plot(sequence, years, 'l') # rearrange time vector
```



```
# Verifying there's no NA's
sum(is.na(temp_carbon)) # [1] 391, missing data.
```

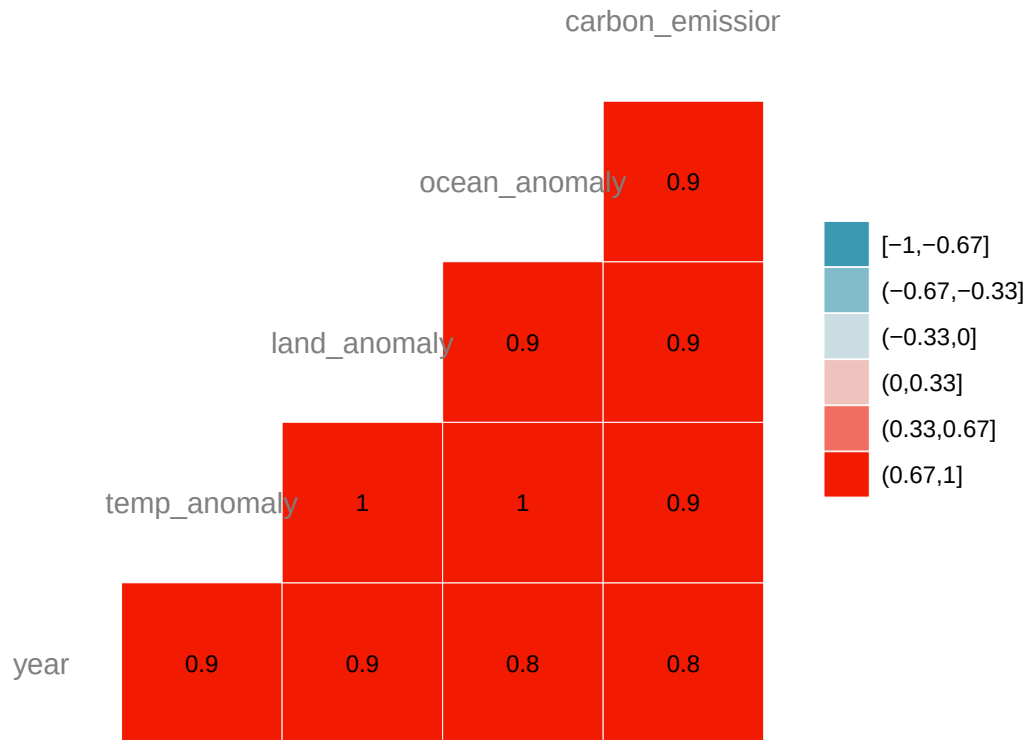
```
## [1] 391
```

```
temp_carbon %>% missmap(col = c("black", "grey"))
```



Checking variable correlations

```
ggcorr(temp_carbon,  
  nbreaks = 6,  
  label = TRUE,  
  label_size = 3,  
  color = "grey50")
```



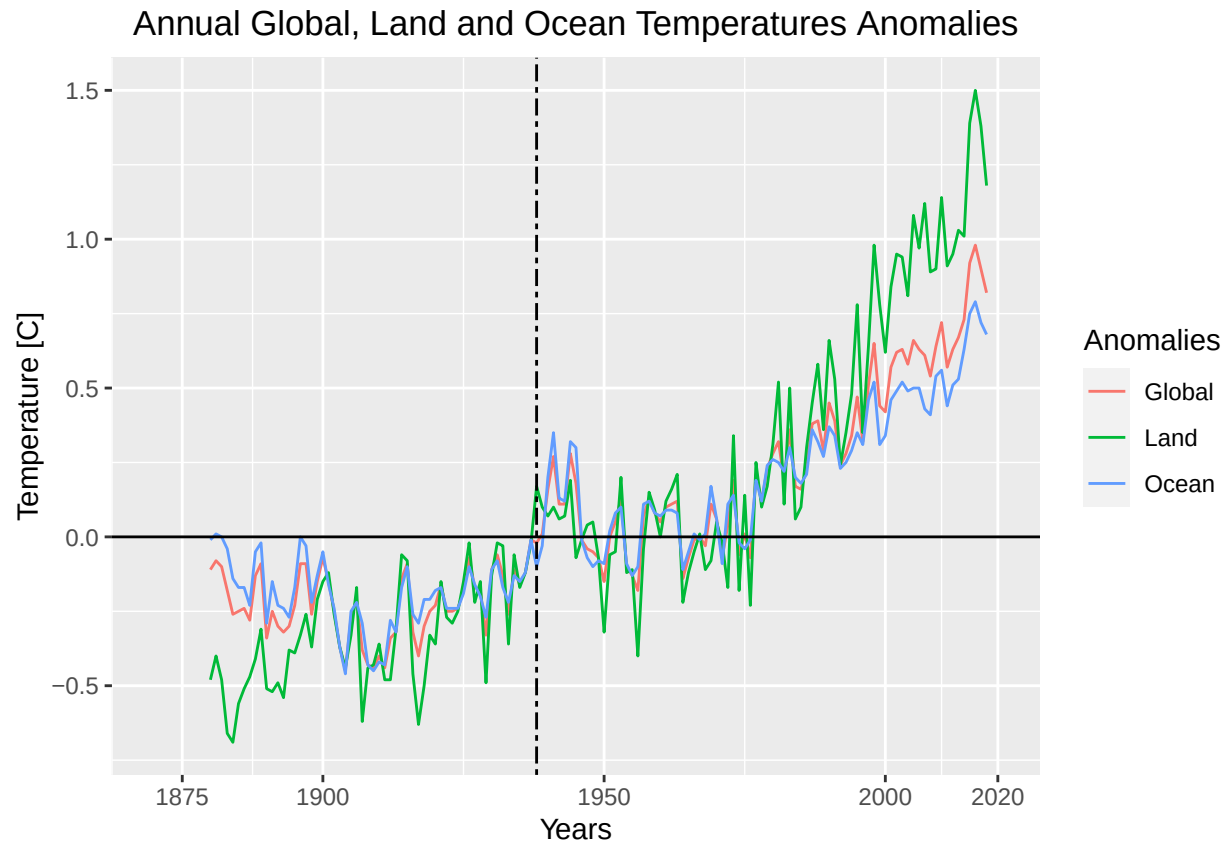
Graphs of Global, Land and Ocean Temperatures Anomalies

```
Global <- temp_carbon$temp_anomaly
Land <- temp_carbon$land_anomaly
Ocean <- temp_carbon$ocean_anomaly

df <- data.frame(years,
                  Global,
                  Land,
                  Ocean)

dfplot <- df %>% gather(key, value, -years)

ggplot(dfplot, mapping = aes(x = years,
                              y = value,
                              color = key)) +
  geom_line(na.rm = TRUE) +
  geom_hline(yintercept = 0) +
  geom_vline(xintercept = 1938, linetype = "twodash") +
  scale_x_continuous(limits = c(1870, 2020),
                     breaks = c(1875, 1900, 1950, 2000, 2020)) +
  scale_color_discrete(name = "Anomalies") +
  xlab("Years") +
  ylab("Temperature [C]") +
  ggtitle("Annual Global, Land and Ocean Temperatures Anomalies") +
  theme(plot.title = element_text(hjust = 0.5))
```



Graph for Annual Carbon Emissions

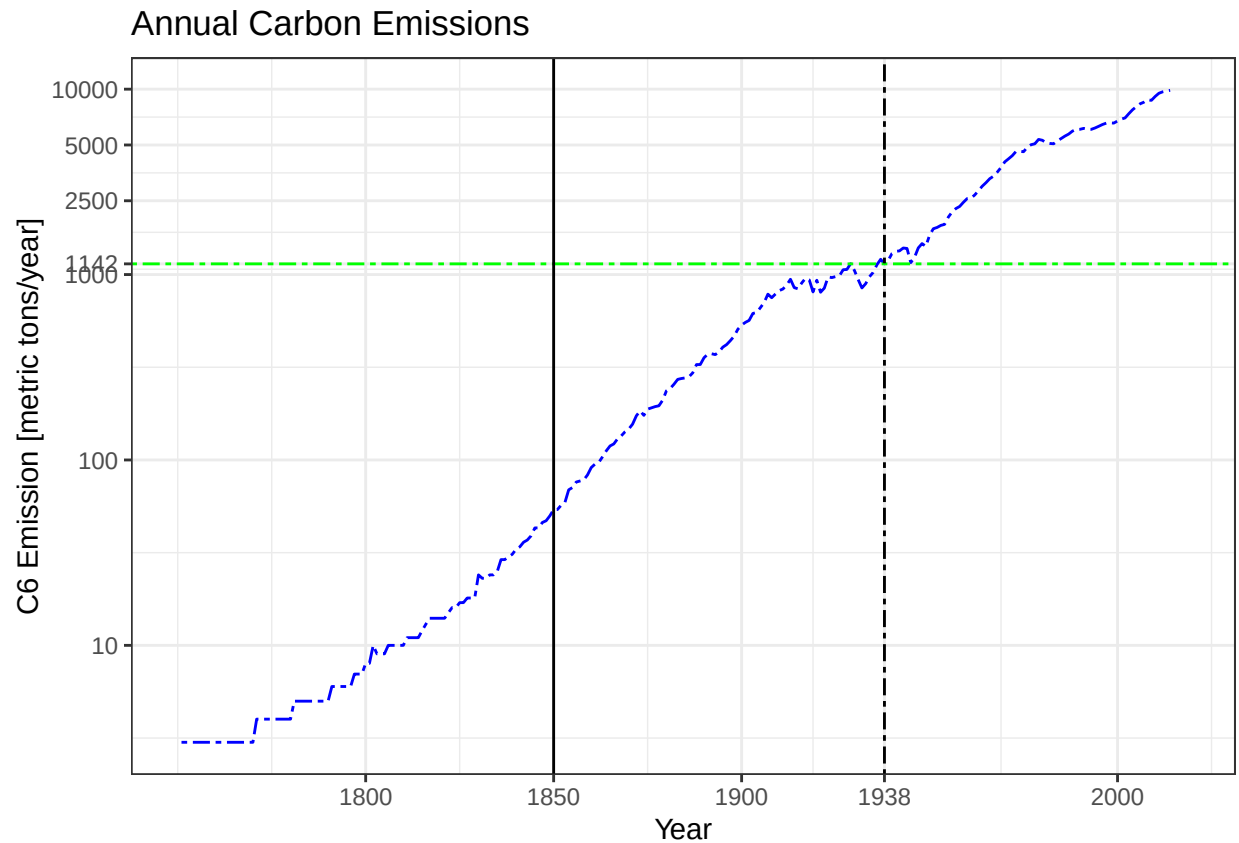
```

yline <- temp_carbon$carbon_emissions[temp_carbon$year == 1938]

temp_carbon %>%
  ggplot(aes(year, carbon_emissions)) +
  geom_line(na.rm = TRUE,
            color = "blue",
            linetype = "twodash") +
  geom_hline(yintercept = 0) +
  geom_hline(yintercept = yline,
            color = "green",
            linetype = "twodash") +
  geom_vline(xintercept = 1850, linetype = "solid") +
  geom_vline(xintercept = 1938, linetype = "twodash") +
  scale_y_log10(breaks = c(0, 10, 100, 1000, yline, 2500, 5000, 10000)) +
  scale_x_continuous(breaks = c(1800, 1850, 1900, 1938, 2000)) +
  xlab("Year") +
  ylab("C6 Emission [metric tons/year]") +
  ggtitle("Annual Carbon Emissions") +
  theme_bw()

```

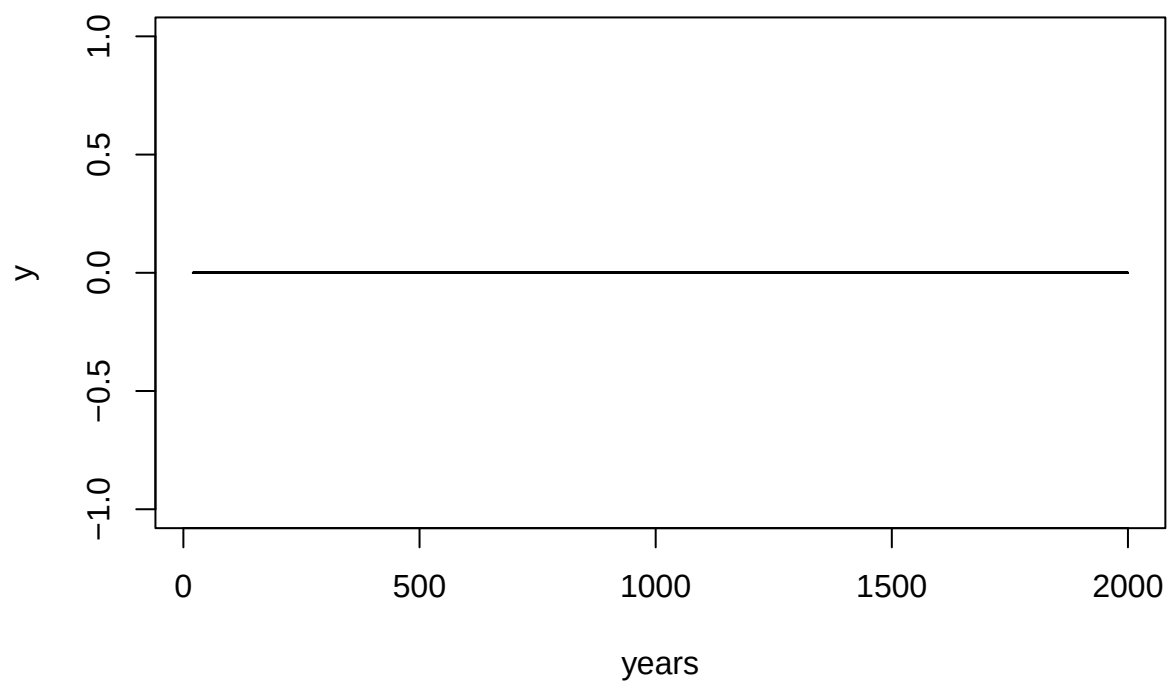
```
## Warning: Transformation introduced infinite values in continuous y-axis
```



2. GREENHOUSE_GASES

Let's check that time vector doesn't have gaps. The time vector (year) has three time lines for each gas. First we run a code for all vectors. If something is wrong we would run it for each gas

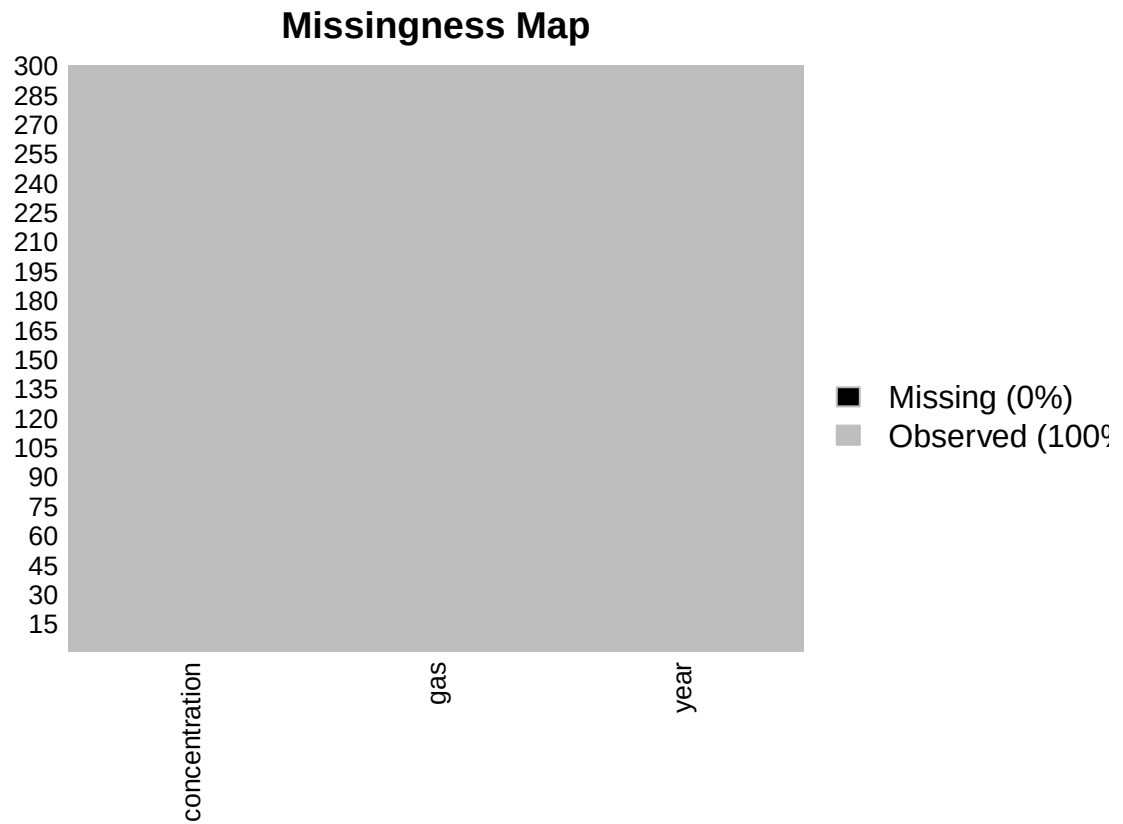
```
years <- greenhouse_gases$year
y <- integer(length(greenhouse_gases$year)) # making a vector of 0's;
plot(years, y, "l") # normalizing with 100 for each gas at the end
```



```
# Verifying that there's no NA's  
sum(is.na(greenhouse_gases))      # [1] 0, missing data.
```

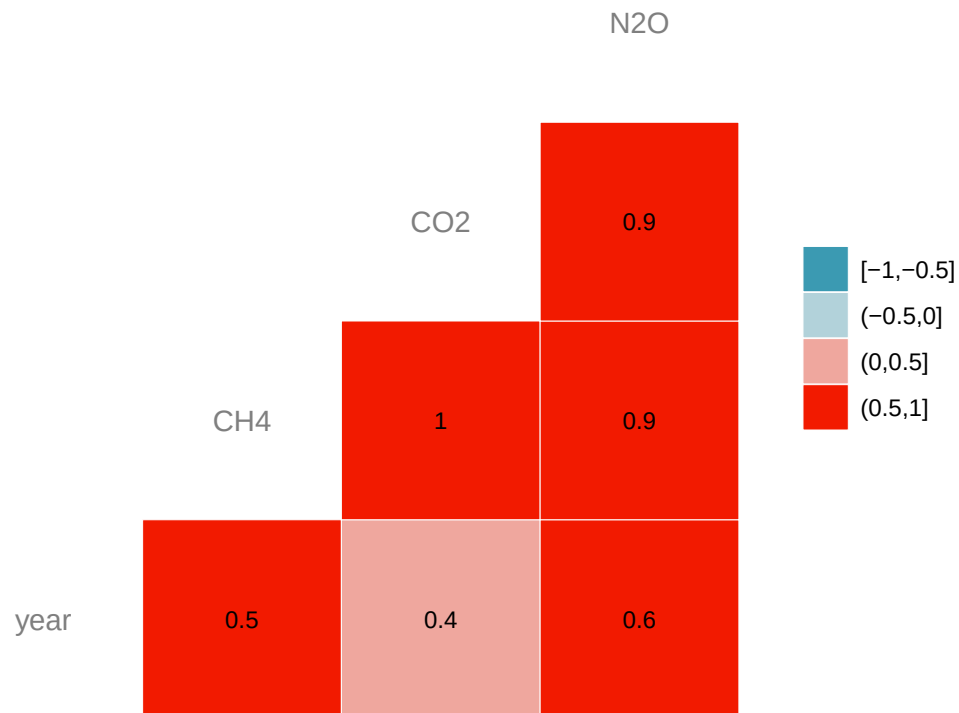
```
## [1] 0
```

```
greenhouse_gases %>% missmap(col = c("black", "grey"))
```

Checking variables correlation

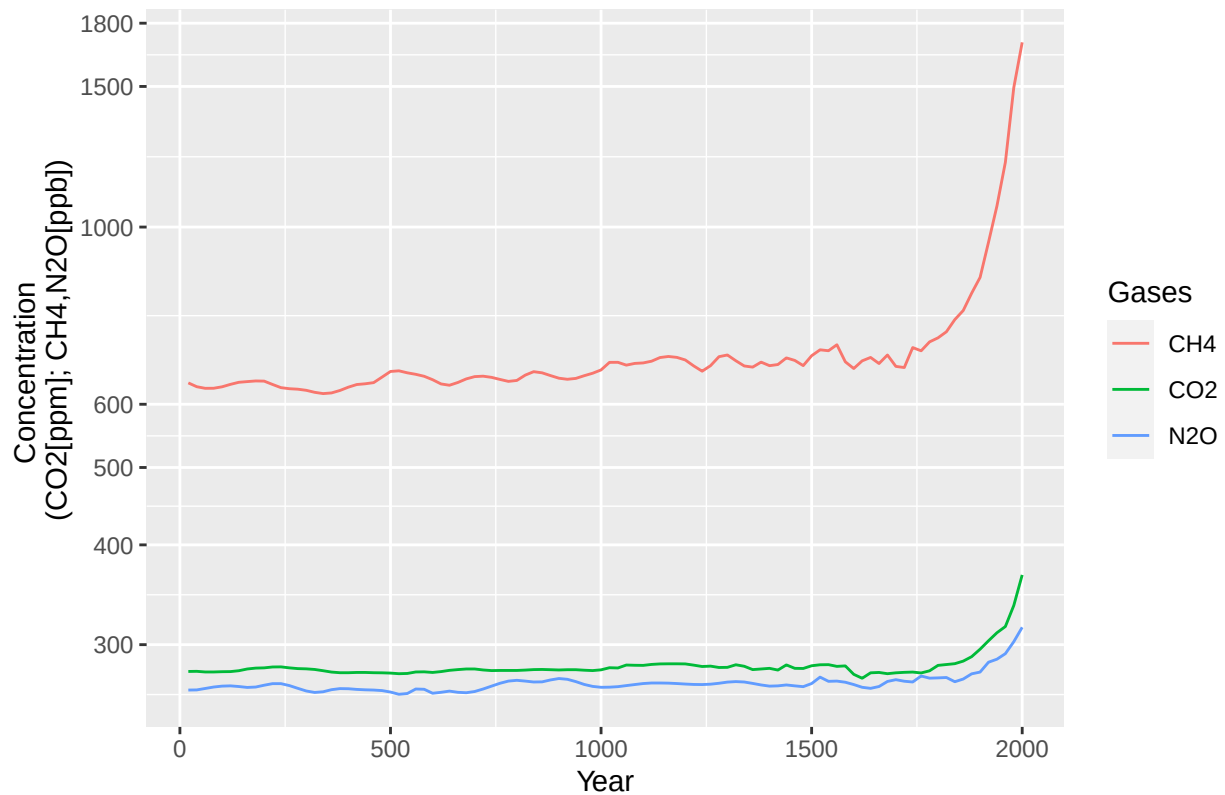
```
df <- greenhouse_gases %>%  
  spread(gas, concentration)  
  
ggcorr(df,  
  nbreaks = 4,  
  label = TRUE,  
  label_size = 3,  
  color = "grey50")
```



Graphs of Concentration of Greenhouse Gases

```
# All
# 1.1
greenhouse_gases %>%
  mutate(gas = factor(gas)) %>%
  group_by(gas) %>%
  ggplot(aes(year, concentration, color = gas)) +
  geom_line() +
  scale_y_continuous(trans = "log10",
                     breaks = c(0, 200, 300, 400, 500, 600, 1000, 1500, 1800)) +
  scale_color_discrete(name = "Gases") +
  xlab("Year") +
  ylab("Concentration
       (CO2[ppm]; CH4,N2O[ppb])") +
  ggtitle("Global Gas Concentration over 2000 years.")
```

Global Gas Concentration over 2000 years.



```
# Each
# 1.2.1 CO2 & N2O
average <- greenhouse_gases %>%
  filter(year %in% seq(0, 1840, 20)) %>%      # 1850 unofficial year it begins
                                              # the industrial revolution

  summarize(CH4 = mean(concentration[gas == "CH4"]),
            CO2 = mean(concentration[gas == "CO2"]),
            N2O = mean(concentration[gas == "N2O"]))

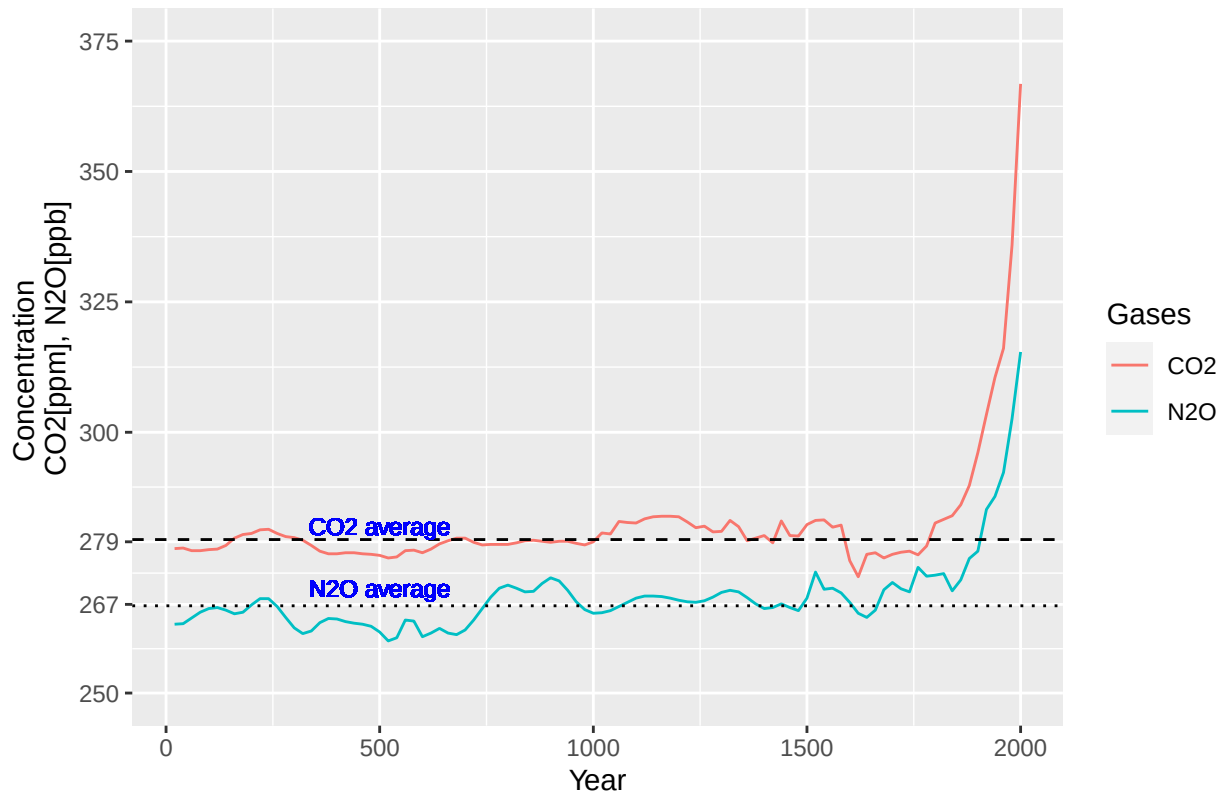
greenhouse_gases %>%
  filter(gas %in% c("CO2", "N2O")) %>%
  mutate(gas = factor(gas)) %>%
  group_by(gas) %>%
  ggplot(aes(year, concentration, color = gas)) +
  geom_line() +
  geom_hline(yintercept = average$CO2, linetype = "dashed") +
  geom_hline(yintercept = average$N2O, linetype = "dotted") +
  scale_y_continuous(limits = c(250, 375),
                    breaks = c(250, 267, 279, 300, 325, 350, 375)) +
  scale_color_discrete(name = "Gases") +
  xlab("Year") +
  ylab("Concentration
       CO2[ppm], N2O[ppb]") +
  ggtitle("Global Gas Concentration of CO2 and N2O over 2000 years.") +
  theme(plot.title = element_text(hjust = 0.5)) +
  geom_text(x = 500, y = 282,
```

```

    label = "CO2 average", colour = "blue", size = 3,
    show.legend = FALSE) +
  geom_text(x = 500, y = 270,
    label = "N2O average", colour = "blue", size = 3,
    show.legend = FALSE)

```

Global Gas Concentration of CO2 and N2O over 2000 years.

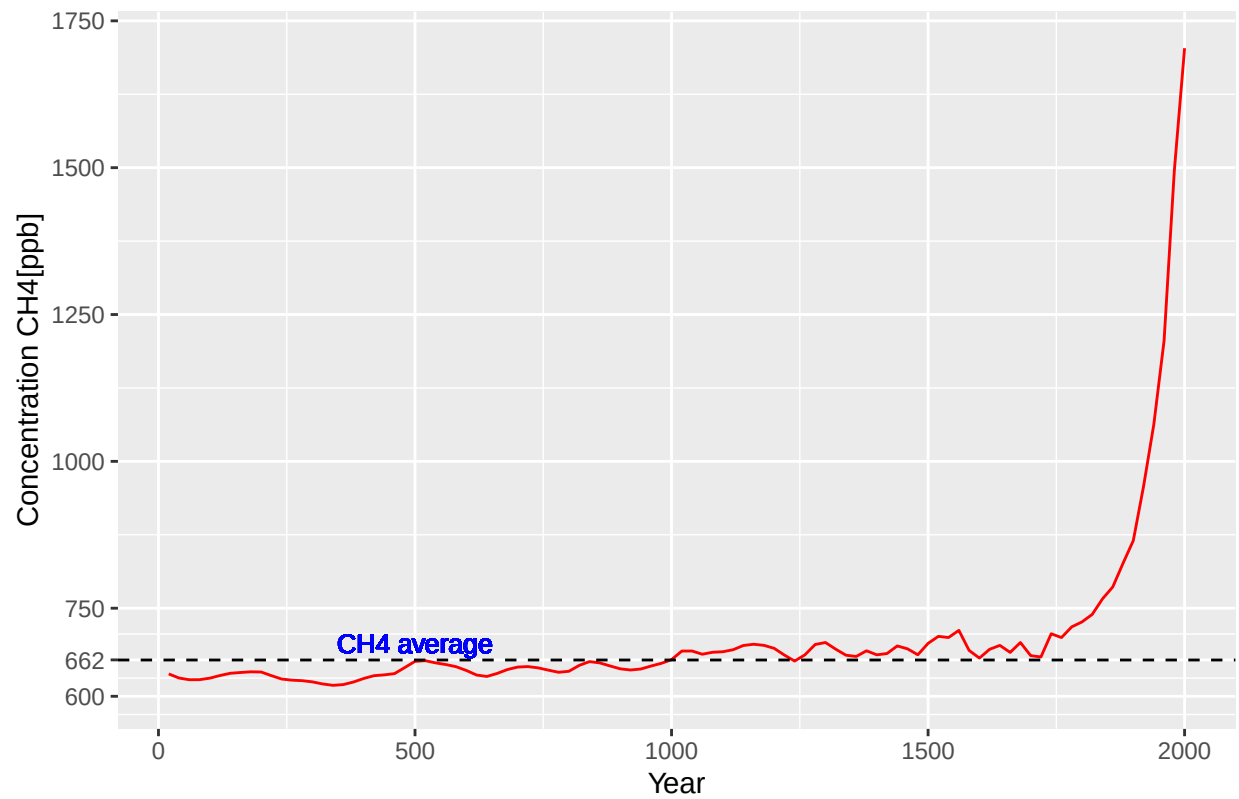


```

# 1.2.2 CH4
greenhouse_gases %>%
  filter(gas == c("CH4")) %>%
  ggplot(aes(year, concentration)) +
  geom_line(color = "red") +
  geom_hline(yintercept = average$CH4, linetype = "dashed") +
  scale_y_continuous(limits = c(600, 1710),
    breaks = c(600, 662, 750, 1000, 1250, 1500, 1750)) +
  xlab("Year") +
  ylab("Concentration CH4[ppb]") +
  ggtitle("Global Gas Concentration of CH4 over 2000 years.") +
  theme(plot.title = element_text(hjust = 0.5)) +
  geom_text(x = 500, y = 690,
    label = "CH4 average", colour = "blue", size = 3.5,
    show.legend = FALSE)

```

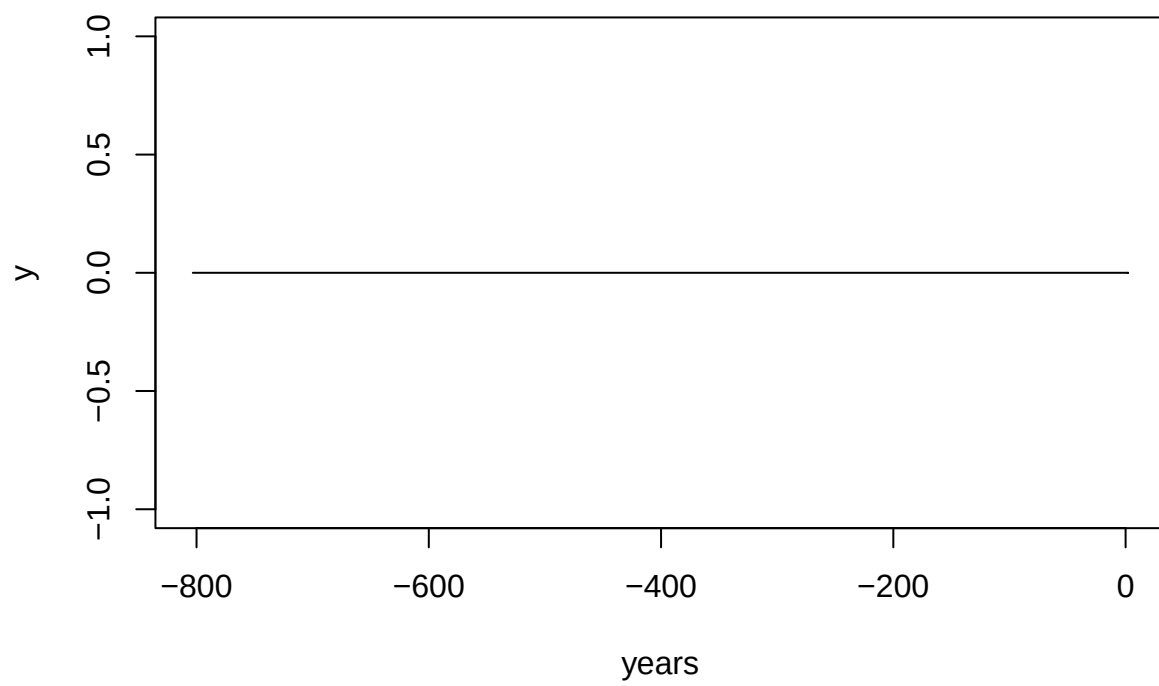
Global Gas Concentration of CH₄ over 2000 years.



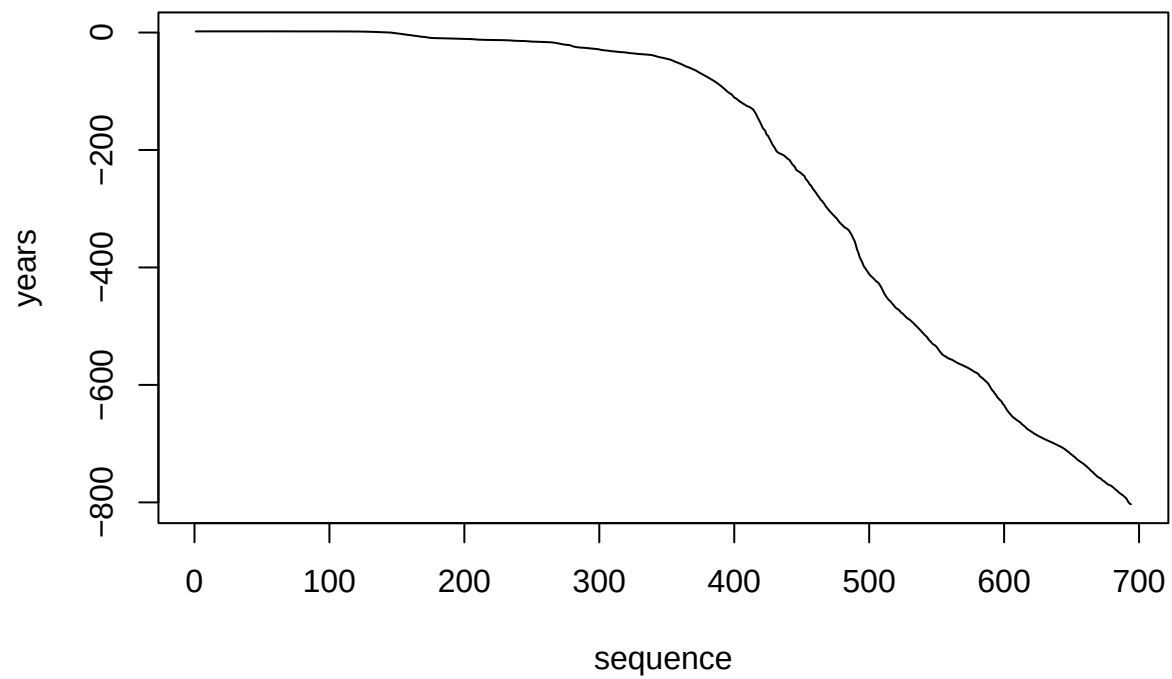
3. HISTORIC_CO2

Checking that time vector doesn't have gaps.

```
years <- historic_co2$year/1000 # time is in thousands years.  
y <- integer(length(historic_co2$year))  
plot(years, y, "l")
```



```
sequence <- seq(1, 694)
plot(sequence, years, 'l')
```



```
# Verifying that there's no NA's
sum(is.na(historic_co2))      # [1] 0, missing data.
```

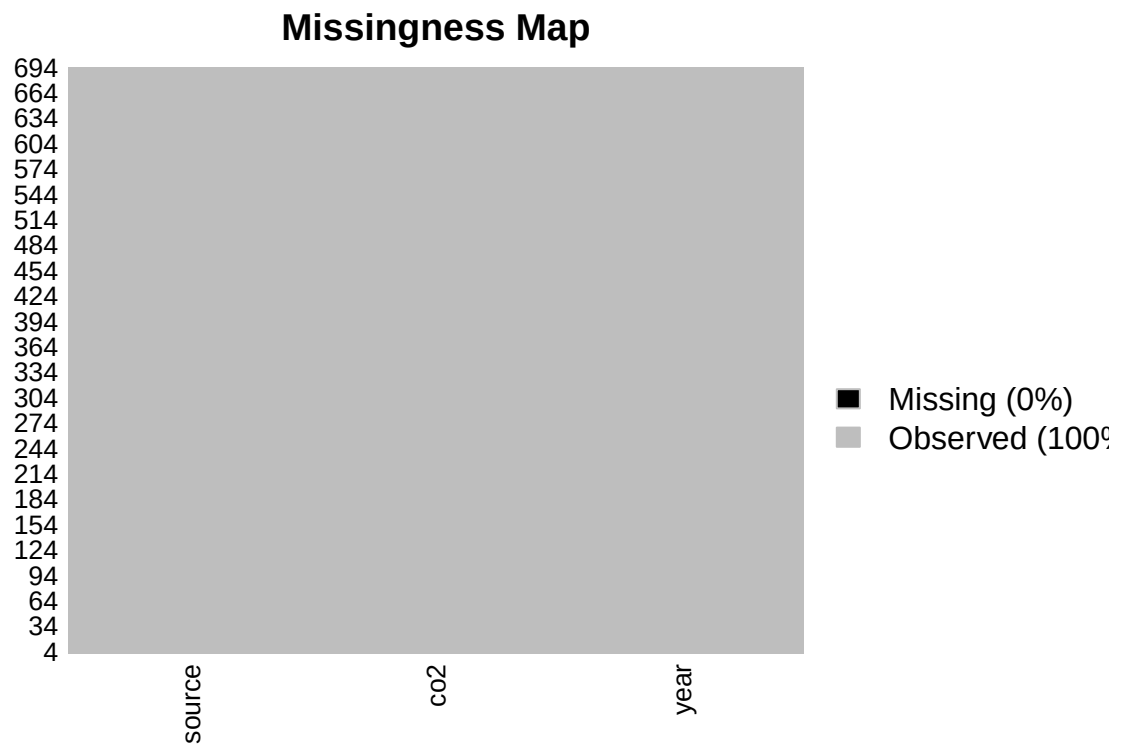
```
## [1] 0
```

```
historic_co2 %>% missmap(col = c("black", "grey"))
```

```
## Warning: Unknown or uninitialised column: 'arguments'.
```

```
## Unknown or uninitialised column: 'arguments'.
```

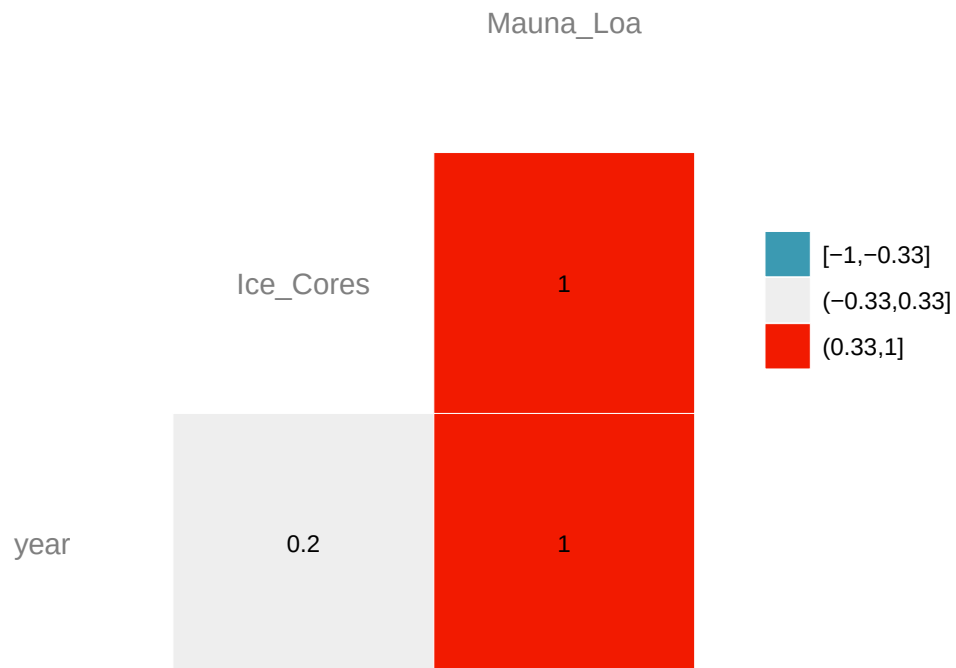
```
## Warning: Unknown or uninitialised column: 'imputations'.
```



Checking variables correlation

```
df <- historic_co2 %>%
  spread(source, co2)

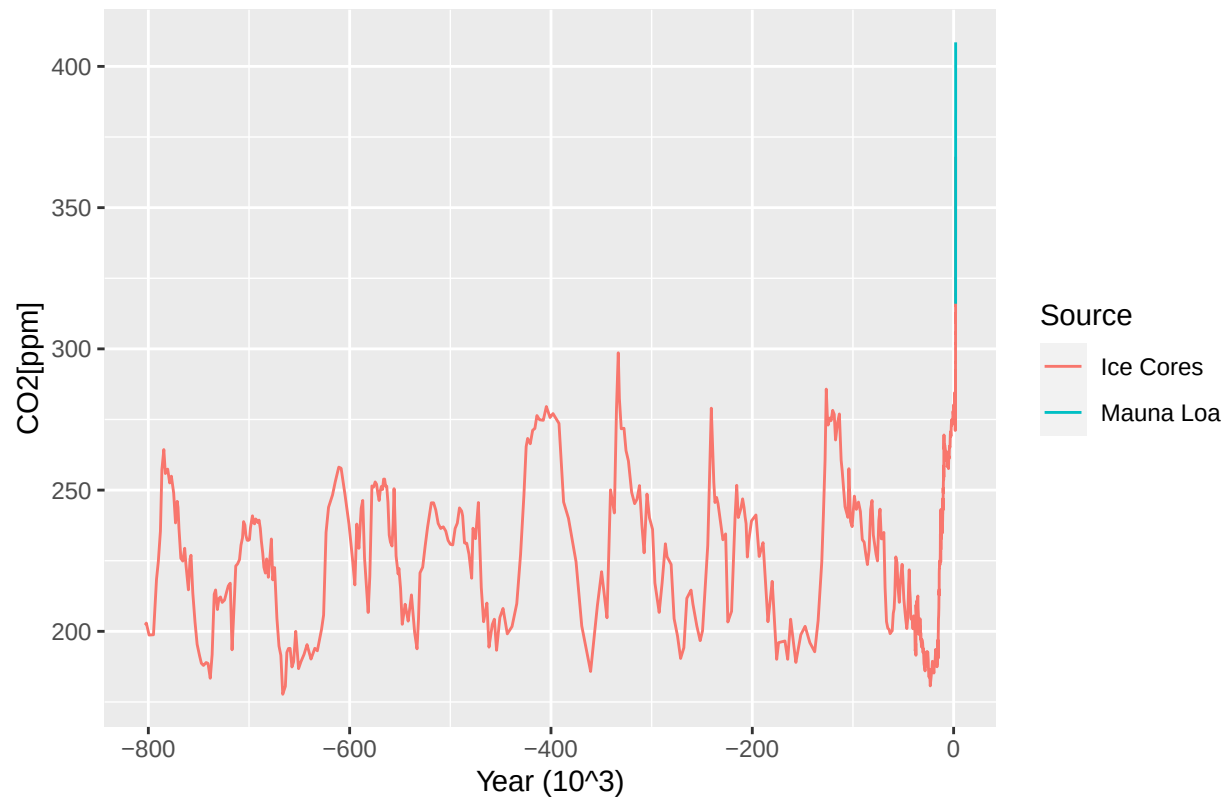
ggcorr(df,
  nbreaks = 3,
  label = TRUE,
  label_size = 3,
  color = "grey50")
```

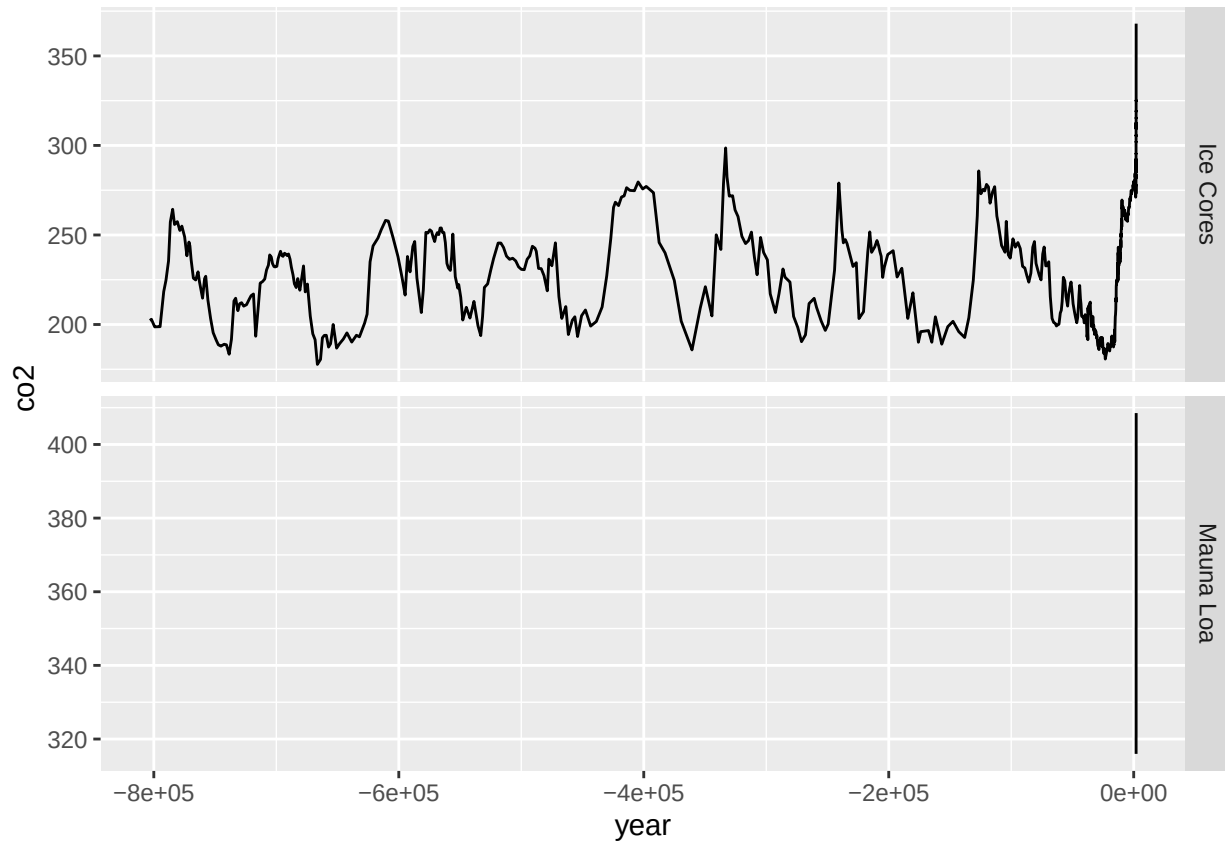
Graphs of Concentration of Historic CO2.

```
# 1.1 Ice Cores and Mauna Loa data in one graph one figure.
historic_co2 %>%
  mutate(source = factor(source)) %>%
  group_by(source) %>%
  ggplot(aes(year/1000, co2, color = source)) +
  geom_line() +
  scale_color_discrete(name = "Source") +
  xlab("Year (10^3)") +
  ylab("CO2[ppm]") +
  ggtitle("Historic CO2 record from > 800,000 years")
```

Historic CO2 record from > 800,000 years



```
# 1.2 Ice Cores and Mauna Loa data in two graphs one figure ('facet_grid').
historic_co2 %>%
  mutate(source = factor(source)) %>%
  group_by(source) %>%
  ggplot(aes(year, co2)) +
  geom_line() +
  facet_grid(source ~ ., scales = "free")
```



```
# 1.3 Ice Cores and Mauna Loa data in two graphs one figure ('grid.arrange').
avg_ic <- historic_co2 %>%
  filter(year < 1850 & source == "Ice Cores") %>%
  summarize(CO2 = mean(co2))

avg_ml <- historic_co2 %>%
  filter(year >= 1850 & source == "Mauna Loa") %>%
  summarize(CO2 = mean(co2))

p1 <- historic_co2 %>%
  filter(source == "Ice Cores") %>%
  ggplot(aes(year/1000, co2)) +
  geom_line(color = "blue") +
  geom_hline(yintercept = avg_ic$CO2) +
  geom_hline(yintercept = 368, linetype = "longdash") +      # 368 ppm, max of CO2 Ice Cores
  geom_vline(xintercept = 1850/10^3, linetype = "dashed") +
  geom_vline(xintercept = 1959/10^3, linetype = "longdash") +
  scale_y_continuous(breaks = c(200, avg_ic$CO2, 250, 300, 350, 368)) +
  xlab("Year*(10^3)") +
  ylab("Ice Cores CO2[ppm]") +
  ggtitle("Historic CO2 record > 800,000 from Ice Cores")

p2 <- historic_co2 %>%
  filter(source == "Mauna Loa") %>%
  ggplot(aes(year, co2)) +
  geom_line(color = "red") +
```

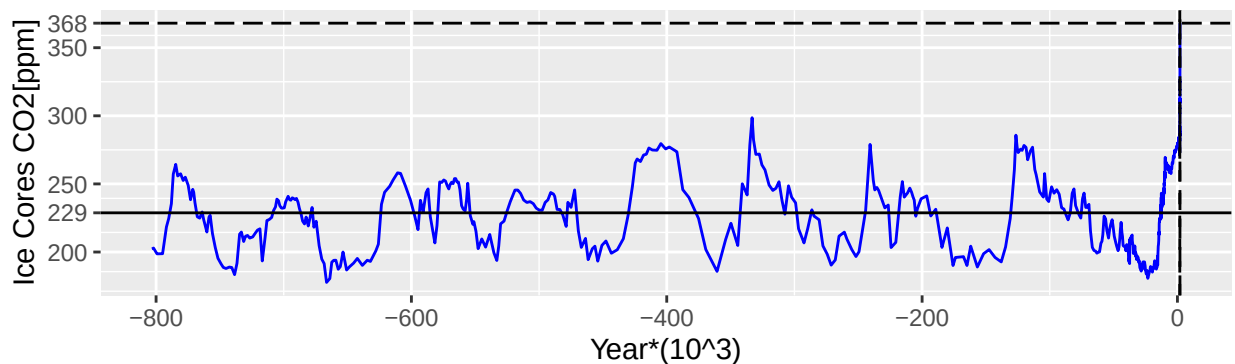
```

geom_point() +
geom_hline(yintercept = avg_ml$CO2) +
geom_hline(yintercept = 368, linetype = "longdash") +
xlab("Year") +
ylab("Mauna Loa CO2[ppm]") +
ggtitle("Historic CO2 record from Mauna Loa") +
geom_text(x = 1970, y = 360,
          label = "CO2[ppm] = 354",
          color = "blue",
          size = 3.5)

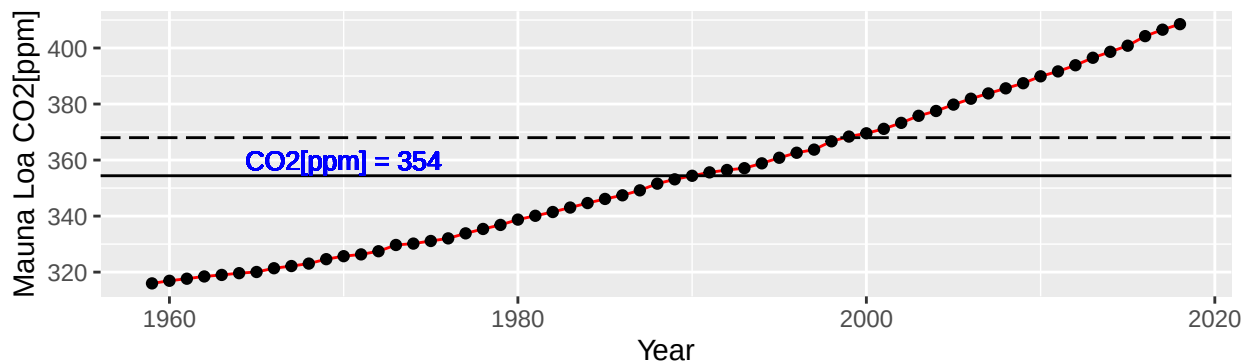
grid.arrange(p1, p2, nrow = 2)

```

Historic CO2 record > 800,000 from Ice Cores



Historic CO2 record from Mauna Loa



```

# 1.4.1 Ice Cores and Grenhouse CO2 data base comparison (one graph).
df_ic <- data.frame(year = historic_co2$year[historic_co2$year > 0 &
                                historic_co2$source == "Ice Cores"],
                    CO2 = historic_co2$co2[historic_co2$year > 0 &
                                historic_co2$source == "Ice Cores"])

df_gh <- data.frame(year = greenhouse_gases$year[greenhouse_gases$gas == "CO2"],
                    CO2 = greenhouse_gases$concentration[greenhouse_gases$gas == "CO2"])

df <- rbind(df_ic, df_gh)

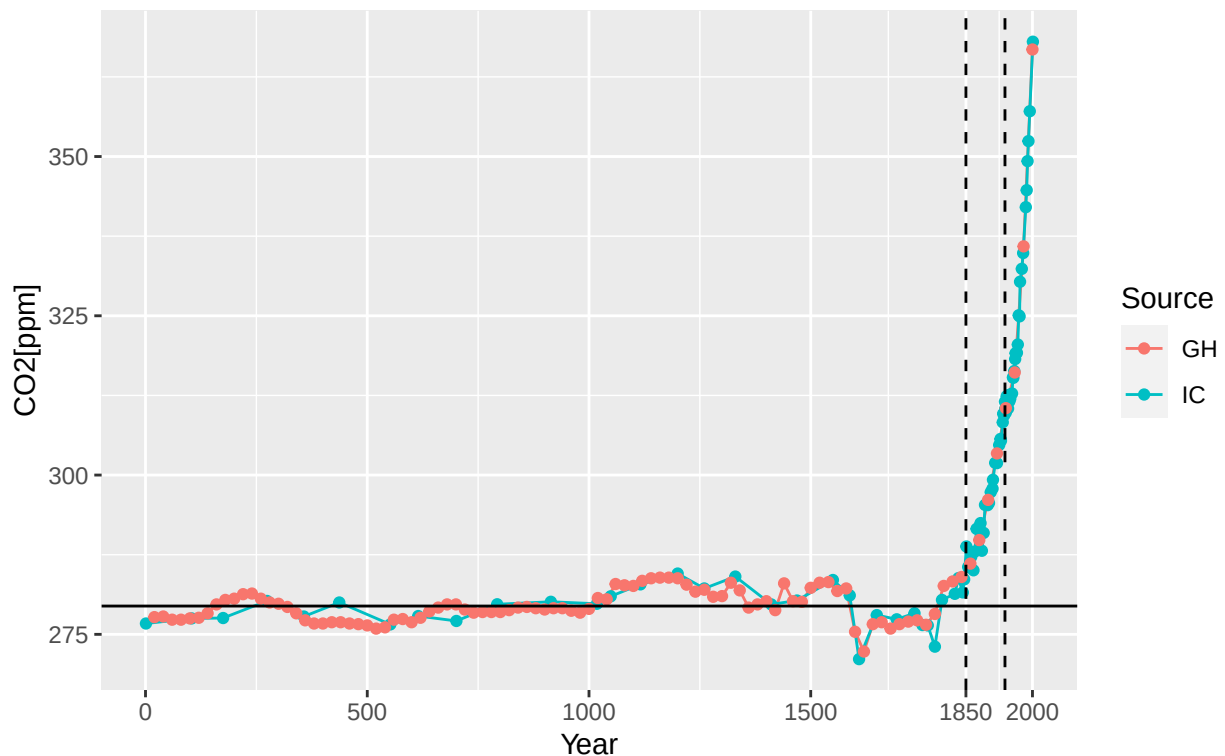
```

```
df$dataset <- c(rep("IC", nrow(df_ic)), rep("GH", nrow(df_gh)))

avg <- (mean(df_ic$CO2[df_ic$year < 1850]) +
       mean(df_gh$CO2[df_gh$year < 1850]))/2

ggplot(data = df, aes(year, CO2, col = dataset)) +
  geom_line() +
  geom_point() +
  geom_hline(yintercept = avg) +
  geom_vline(xintercept = 1850, linetype = "dashed") +
  geom_vline(xintercept = 1938, linetype = "dashed") +
  scale_x_continuous(breaks = c(0, 500, 1000, 1500, 1850, 2000)) +
  xlab("Year") +
  ylab("CO2[ppm]") +
  scale_color_discrete(name = "Source") +
  ggtitle("Comparison between CO2 from Greenhouse gases(GH)
          and Ice Cores(IC)")
```

Comparison between CO2 from Greenhouse gases(GH)
and Ice Cores(IC)



```
# 1.4.2 (two graph)
p <- historic_co2 %>%
  filter(year > 0 & source == "Ice Cores") %>%
  ggplot(aes(year, co2)) +
  geom_line() +
  geom_point() +
  geom_vline(xintercept = 1850, linetype = "dashed") +
```

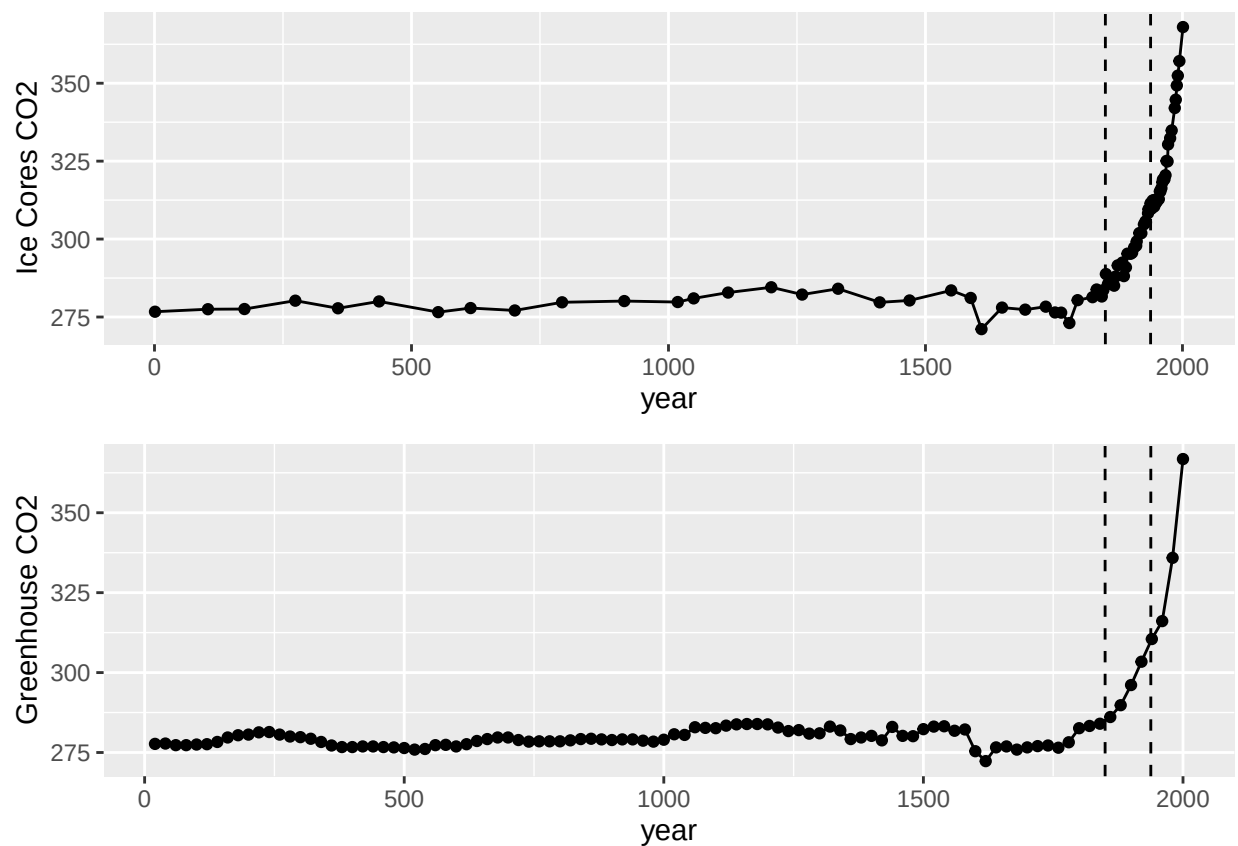
```

geom_vline(xintercept = 1938, linetype = "dashed") +
ylab("Ice Cores CO2")

q <- greenhouse_gases %>%
  filter(gas == "CO2") %>%
  ggplot(aes(year, concentration)) + # length 100
  geom_line() +
  geom_point() +
  geom_vline(xintercept = 1850, linetype = "dashed") +
  geom_vline(xintercept = 1938, linetype = "dashed") +
  ylab("Greenhouse CO2")

grid.arrange(p, q, nrow = 2)

```



```

# 1.5 Ice Cores and Mauna Loa data from 1959.
historic_co2 %>%
  filter(year >= 1959) %>%
  ggplot(aes(year, co2, color = source)) +
  geom_line() +
  geom_point() +
  geom_vline(xintercept = 1959, linetype = "dashed") +
  xlab("Year") +
  ylab("CO2[ppm]") +
  ggtitle("CO2 from Ice Cores and Mauna Loa")

```

CO2 from Ice Cores and Mauna Loa

